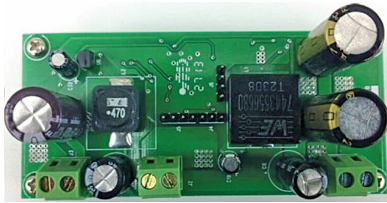


The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.

Solar Powered Streetlight



This reference design is for a self-sufficient solar charger controller that can drive LEDs without any external power source. The reference design features Maximum Power Point Tracking (MPPT) functionality that can charge a 12V battery pack and drive up to 15 LEDs in series.

This reference design from Texas Instruments is of a high-efficiency solar-powered streetlight. The reference design integrates a 12A maximum power point tracking (MPPT) solar charge controller with a 700mA LED driver. The reference design measures only 100mm x 45mm x 32mm. It has an efficiency of over 90% for the LED driver and over 95% for the MPPT charger. The streetlight reference design can drive up to 15 LEDs in series. The design works on a 12V solar panel but by changing the MOSFETs, it is compatible with a higher rated solar panel of up to 24V. The design can charge various chemistries of batteries. The design is made for a 12V Li-ion battery, but with minor modifications, it can be used for a 12V lead-acid battery or a 24V Li-ion battery system by just changing the MOSFETs. The reference design can effectively convert power from a solar system with MPPT, safely charging various battery chemistry types and accurately controlling power flow. It combines safety features such as reverse battery protection and comes with a built-in battery profile for a lead-acid battery, thus, reducing the BoM components and cost. Furthermore, this design can speed up the development phase, resulting in a faster time to market.

- Solar charge controller
- Solar streetlight

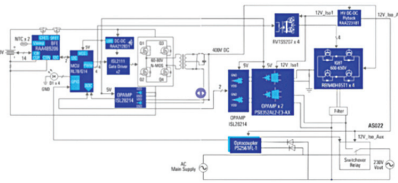
Texas Instruments

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-solar-powered-street-light>

https://qrco.de/solar_light



Sine Wave Inverter



This reference design is of a general-purpose inverter which can be used both as a high-frequency inverter or a low-frequency inverter. The reference design also features a multi-cell battery management IC that can optimise battery life and prevent catastrophic failures.

The Renesas reference design is of a pure sine wave inverter with a capacity (rating) range from 800VA to 1800VA (based on the battery size). This inverter design can provide an output of 230V AC at a frequency of 50Hz/60Hz, generated using the battery pack. The same design can be utilised for building both a low-frequency and a high-frequency inverter. The safety of the battery in the reference design is controlled by an intelligent battery management system (BMS) that employs a 16-cell battery front-end (BFE) IC, which can protect the battery pack and the inverter. The BFE periodically scans the battery status and the operating environment to optimise battery life and prevent catastrophic failures. The design also features an LCD which shows various battery performance statistics. The reference employs a microcontroller (MCU) running at a frequency of 32MHz. The MCU offers multiple built-in safety features that can detect the illegal operation of hardware. Moreover, it can enable support for the household appliance safety standard (IEC/UL 60730). The reference design offers an isolated system to track the AC output and features an auxiliary flyback power supply for the isolated system.

- Home inverter
- General inverter
- Battery management system

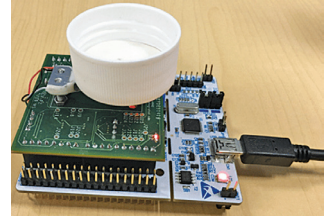
Renesas Electronics

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-of-sine-wave-inverter>

<https://qrco.de/Inverter>



Smart Pill Bottle



This smart pill bottle can detect the consumption of pills by analysing the weight of the pills in the bottle. It will enable patients and healthcare workers to improve the tracking and logging of medicine intake.

The MAXREFDES1150 from Maxim Integrated is a reference design for a smart pill bottle. The design incorporates a weight scale and utilises the Internet of Things (IoT) to monitor and log the real-time medicine intake of a patient. The main features of this reference design are its high accuracy, low power consumption, compatibility with the Mbed Platform and Bluetooth connectivity. MAXREFDES1150 is suitable for a wide variety of IoT measurement applications requiring accurate analogue-signal measurement and communication. The reference design also includes an Android APK with HMI technology. The APK can be used to configure the pills parameter and be updated if needed. The reference design has a container where the patient needs to store the pills. It then allows the user to specify the number of pills and frequency and amount of each dosage. This design is capable of validating the timely consumption of pills based on the weight of pills present in the container. The design also includes an audible buzzer if the smart pill bottle is left on its side or upside down. The reference design can alert the patient and the healthcare staff about the quantity of the medicine for timely replenishment of the depleted medicine.

- Pill container
- Medicine logger

Maxim Integrated

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-smart-pill-bottle>

https://qrco.de/smart_pill_box

